

3. Laplace Transforms

Laplace transform simply takes a calculus problem in the time domain and converts it into an algebra problem in the frequency domain.

$$1. \mathcal{L}\{k\} = \frac{k}{s}$$

$$2. \mathcal{L}\{e^{at}\} = \frac{1}{s-a}$$

$$3. \mathcal{L}\{\cos(\omega t)\} = \frac{s}{s^2 + \omega^2}$$

$$4. \mathcal{L}\{\sin(\omega t)\} = \frac{\omega}{s^2 + \omega^2}$$

$$5. \mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$$

Properties:

1. Linearity

$$\mathcal{L}\{a f(t) + b g(t)\} = a F(s) + b G(s)$$

2. Shift

$$\mathcal{L}\{e^{at} f(t)\} = F(s-a)$$

3. Derivative

$$\mathcal{L}\{f'(t)\} = sF(s) - f(0)$$

$$\mathcal{L}\{f''(t)\} = s^2 F(s) - sf(0) - f'(0)$$

3. Solving initial value problems.

- 1st-order diff eq. of the form

$$\dot{x}(t) + ax(t) = b$$

- Initial condition:

$$x(0) = c$$

1) Translate to freq. domain

$$\dot{x}(t) \rightarrow sX(s) - x(0)$$

$$X(t) \rightarrow X(s)$$

$$b \rightarrow \frac{b}{s}$$

2) Isolate the system variable $X(s)$.

- Move initial condition over to the right hand side $x(0)$

- Group the other terms $X(s)$ on the left hand side:

$$(s+a)X(s)$$

- Combine the right hand side terms into a single, unified fraction

- Divide both sides by $(s+a)$ to fully isolate $X(s)$

3) Partial fraction decomposition: solve the system for A & B , or use the Heaviside cover-up shortcut:

$$X(s) = \frac{s+8}{s(s+2)} = \frac{A}{s} + \frac{B}{s+a}$$

To find A:

- Look at the denominator under A. Cover up the s in the original fraction and substitute $s=0$ into what remains.

$$A = \frac{s+8}{(s+2)} \Big|_{s=0} = \frac{0+8}{0+2} = 4$$

To find B:

- Look at the denominator under B ($s+2=0 \Rightarrow s=-2$). Cover up the $(s+2)$ and substitute $s=-2$ into what remains.

$$B = \frac{s+8}{s} \Big|_{s=-2} = \frac{-2+8}{-2} = \frac{6}{-2} = -3.$$

4) Translate back into the time domain.

Use the inverse (look-up) Laplace:

$$\frac{A}{s} \rightarrow A$$

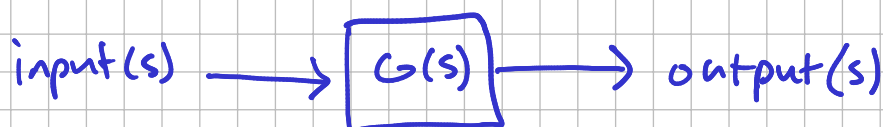
$$\frac{B}{s+a} \rightarrow B e^{-at}$$

Concept: Transfer Function (TF)

In system dynamics and automation, a TF $G(s)$ is an algebraic blueprint of a physical system's internal mechanics.

$$\text{Output}(s) = G(s) \times \text{input}(s)$$

$$\Rightarrow G(s) = \frac{\text{Output}(s)}{\text{Input}(s)}$$



Input: $U(s)$

Output: $Y(s)$ (or $T(s)$)

Strategy: Finding a TF from an Ordinary Diff Eq. (ODE)

1. Clear the starting energy (set initial conditions to zero)
 $x(0) = 0$

2. Swap components for their s counterparts

Input function $u(t) \rightarrow U(s)$

Output variable $y(t) \rightarrow Y(s)$

First derivative $\dot{y}(t) \rightarrow sY(s)$

2nd derivative $\ddot{y}(t) \rightarrow s^2 Y(s)$

3. Isolate and divide to form $G(s)$

- Group everything with $Y(s)$ on the left side and factor out $Y(s)$

- Group all input terms on the right side ($U(s)$)

- Isolate the final target fraction

$$G(s) = \frac{Y(s)}{U(s)}$$

Concept: Poles and zeros

Every transfer function $G(s)$ is structured as a fraction of polynomials. The roots of these determine the algebraic boundaries of how a physical system behaves.

$$G(s) = \frac{N(s)}{D(s)} = \frac{\text{Numerator (zeros)}}{\text{Denominator (poles)}}$$

Zeros: these dictate the strength and amplitude of the response components.

Poles: these dictate the speed, timing, and structural stability of the response.

Strategy: Finding and Classifying Poles and zeros

To analyze any TF:

1. Calculate the zeros

$$N(s) = 0 \Rightarrow s = z_1, z_2, \dots$$

2. Calculate the Poles:

$$D(s) = 0 \Rightarrow s = p_1, p_2, \dots$$

If a factor repeats (example: $(s+1)^2$), the pole has a multiplicity equal to the power (2).

3. Classify the natural system response

- Single real negative pole ($s = -a$): non-oscillating decay
- Complex conjugate poles with negative, real parts ($s = -\sigma \pm j\omega$): decaying oscillation
- Purely imaginary poles ($s = \pm j\omega$): pure oscillation $\Rightarrow$ goes on forever.

Concept: First-Order System Metrics

$\Rightarrow$ First-order diff eqs track systems without any directional inertia

Strategy: Convert a diff eq like

$$a_1 \dot{y}(t) + a_0 y(t) = b_0 r(t)$$

... to the s-domain:

$$(a_1 s + a_0) Y(s) = b_0 R(s).$$

$\Rightarrow$ Divide the entire equation by the constant a_0 to isolate the standard form

$$\left(\frac{a_1}{a_0} s + 1 \right) Y(s) = \frac{b_0}{a_0} R(s)$$

$$\Rightarrow (\tau s + 1) Y(s) = K_{dc} R(s)$$

- Parameter extraction:

$$\tau = \frac{a_1}{a_0}, \quad \text{Static DC gain } K_{dc} = G(0) = \frac{b_0}{a_0}$$

- Step response curve: $R(s) = \frac{M}{s}$

$\Rightarrow$ For an incoming step input of magnitude M
where $r(t) = M u(t)$,

the real-time solution is derived from partial fractions (see Heavyside shortcut above):

$$y(t) = M \cdot K_{dc} (1 - e^{-t/\tau}) u(t)$$

- Final value shortcut:

The long-run value is $y_{\infty} = M K_{dc}$